Math 55—Fall 2025—Haiman
Homework 1
Due on Gradescope Wednesday, Sept. 3, 8pm

Problems from Hutchings notes on Introduction to Mathematical Arguments

Section 2 Exercises (page 12):

1(a,b).
2.
3.
4.

Problems from the Rosen textbook

Section 1.1:

17(c,f) Are these two propositions logically equivalent? Why or why not? (Odd numbered problems have answers in the book, so my follow-up question is the point of this problem.)

Section 1.3:

12 and 14 for 12(b,d). Also, explain how each of these tautologies could be used in a logical argument. As an example of what I mean, an answer for 12(a) could be: "If we know that at least one of two statements is true, and we can rule out the first, then we can conclude that the second is true."

Section 1.4:

46.
54(a,c,d), and explain your answers.

Section 1.5:

24(d).
40(b). Justify your answer.